

Mo Sha

Ph.D. Candidate in Computer Science

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SELECTED PUBLICATIONS

- **Self-adaptive Graph Traversal on GPUs.**
M Sha, Y Li, and KL Tan.
International Conference on Management of Data (**SIGMOD**), 2021.
- **GPU-Accelerated Subgraph Enumeration on Partitioned Graphs.**
W Guo, Y Li, M Sha, B He, X Xiao, and KL Tan.
International Conference on Management of Data (**SIGMOD**), 2020.
- **GPU-based Graph Traversal on Compressed Graphs.**
M Sha, Y Li, and KL Tan.
International Conference on Management of Data (**SIGMOD**), 2019.
- **Accelerating Dynamic Graph Analytics on GPUs.**
M Sha, Y Li, B He, and KL Tan.
International Conference on Very Large Data Bases (**VLDB**), 2018.
- **Parallel Personalized Pagerank on Dynamic Graphs.**
W Guo, Y Li, M Sha, and KL Tan.
International Conference on Very Large Data Bases (**VLDB**), 2018.

PATENT

- **Cross-section extraction for vortex detection.**
M Sha, J Liu, Y Ke, E Leonardi, and K Menzies.
Rolls-Royce, US20190303515A1/EP3547174A1, filed on Mar 2019.

HONORS & AWARDS

- Dean's Graduate Research Excellence Award, NUS, 2020.
- Research Achievement Award, NUS, 2018.
- First Prize, ACM/ICPC Asia Regional, Hangzhou, 2012.
- The CCF Outstanding Undergraduate Award, Dalian, 2012.

PROFESSIONAL SERVICES

External Reviewer:

- International Conference on Very Large Data Bases, 2021.
- IEEE International Conference on Data Engineering, 2019.

Invited Reviewer:

- Journal of Parallel and Distributed Computing, 2019.
- IEEE Access, 2019.

TEACHING

Teaching Assistant:

- CS5234: Algorithm at Scale, NUS, 2019.
- CS3223: Database Systems Implementation, NUS, 2018-2021.
- CS1101S: Programming Methodology, NUS, 2018.

EDUCATION

Ph.D. in Computer Science

supervised by: [Prof. Kian-Lee Tan](#)

National University of Singapore

Aug 2017 – present Singapore

B.Eng. in Computer Science

Harbin Institute of Technology

Sep 2010 – Jun 2014 Harbin, China

EXPERIENCES

Research Assistant

Database Lab, School of Computing, NUS

Jun 2016 – Jun 2017 Singapore

Our team was interested in dynamic graph analytics on various applications and was exploring to utilize different hardware hierarchies, e.g., GPU accelerators and their combined architecture, for higher efficiency.

Project Officer

Rolls-Royce@NTU Corporate Lab, NTU

Jun 2015 – Jun 2016 Singapore

I took part in the research project focusing on the automatic vortex extraction and tracking in CFD data, which is an essential tool for the aerodynamic design of Rolls-Royce's jet engine.

Algorithm Engineer

Alibaba

Jun 2014 – Jun 2015 Beijing, China

I was responsible for the backend of the business logic layer of internet advertisement delivery engine, which was mainly related to user profiling and targeted advertisement indexing, retrieving and recalling.

Software Engineering Intern

Google

Jun 2013 – Nov 2013 Shanghai, China

Our team was in charge of developing the backend transaction logic management of the Demand Side Platform, and my main contribution during the internship was the RetryAction module.